Microsoft Fabric meets MCP – what can we do with it

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With great power comes great responsibility.

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FAVORITE STUFF:

Microsoft Fabric

The unified data platform for AI transformation

- Data Factory
- Analytics
- Databases
- Real-Time Intelligence
- IQ
- Power BI

Fabric Platform

- Copilot
- OneLake
- Governance

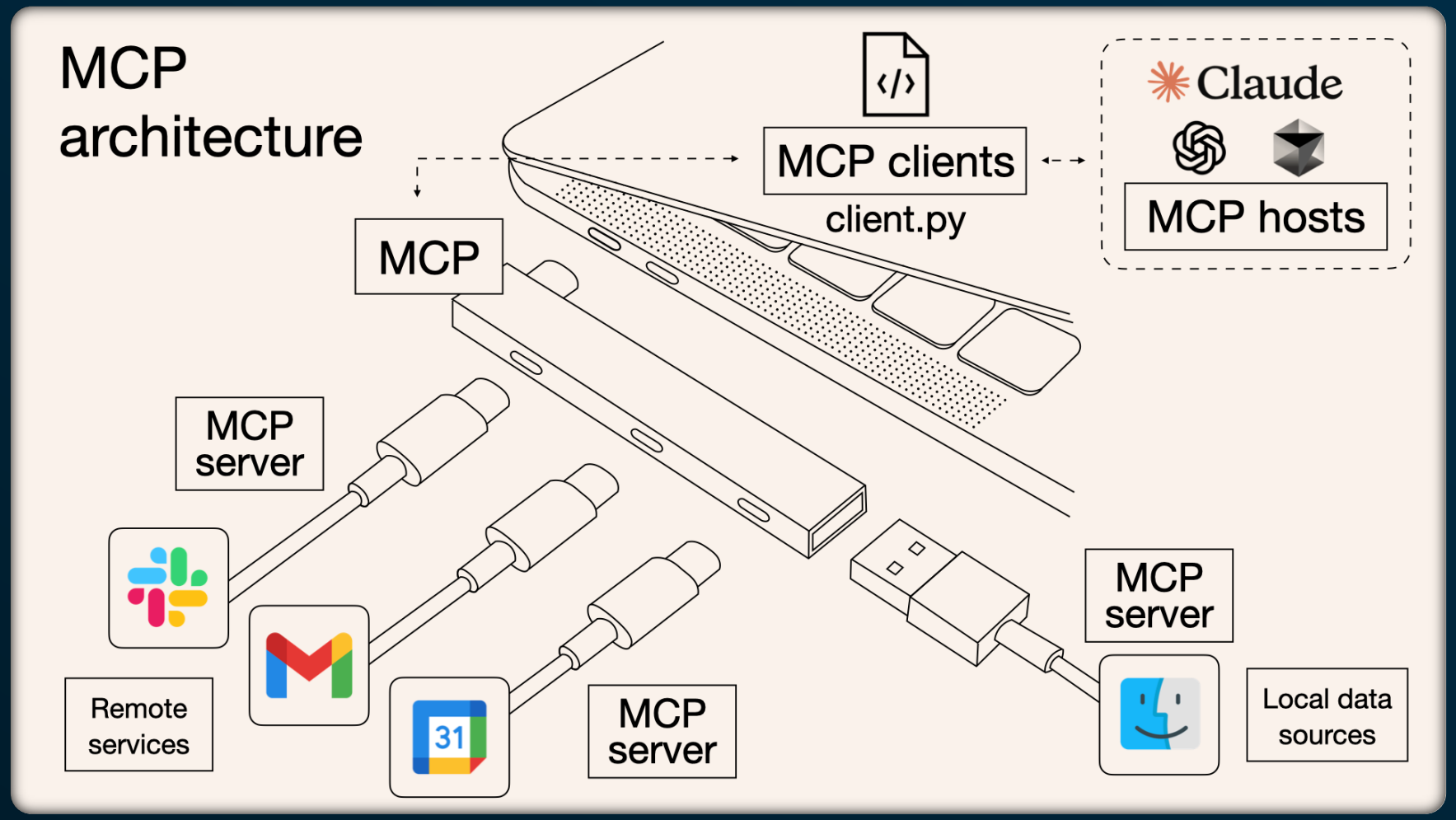
What is MCP?

MCP (Model Context Protocol) is an open protocol designed to standardize how applications provide context to Large Language Models (LLMs).

MCP was started as a project by Anthropic to make it easier for AI models - like Claude - to interact with tools and data sources.

Learn more: https://modelcontextprotocol.io

MCP Architecture



Source: Norah Sakal's blog - What is Model Context Protocol (MCP)? How it simplifies AI integrations compared to APIs | AI Agents That Work

MCP Glossary

- **MCP Hosts:** These are applications (like Claude Desktop or AI-driven IDEs) needing access to external data or tools
- **MCP Clients:** They maintain dedicated, one-to-one connections with MCP servers
- **MCP Servers:** Lightweight servers exposing specific functionalities via MCP, connecting to local or remote data sources
- **Local Data Sources:** Files, databases, or services securely accessed by MCP servers
- **Remote Services:** External internet-based APIs or services accessed by MCP servers

Fabric & Power BI MCP Servers by Microsoft

- **Microsoft Fabric MCP Server**
  https://github.com/microsoft/mcp/tree/main/servers/Fabric.Mcp.Server

- **Fabric Admin MCP Server**
  https://github.com/microsoft/fabric-admin-mcp

- **Power BI MCP Modeling Server**
  https://github.com/microsoft/powerbi-modeling-mcp

- **Microsoft Fabric RTI MCP Server**
  https://github.com/microsoft/fabric-rti-mcp

- **Microsoft Data Factory MCP Server** NEW
  https://github.com/microsoft/DataFactory.MCP

Fabric Admin MCP Server

Use Cases

- 📄 List Fabric capacities
- 📑 Get capacity details
- ⏸️ Pause capacity
- ▶️ Resume capacity

Microsoft Fabric MCP Server

Use Cases

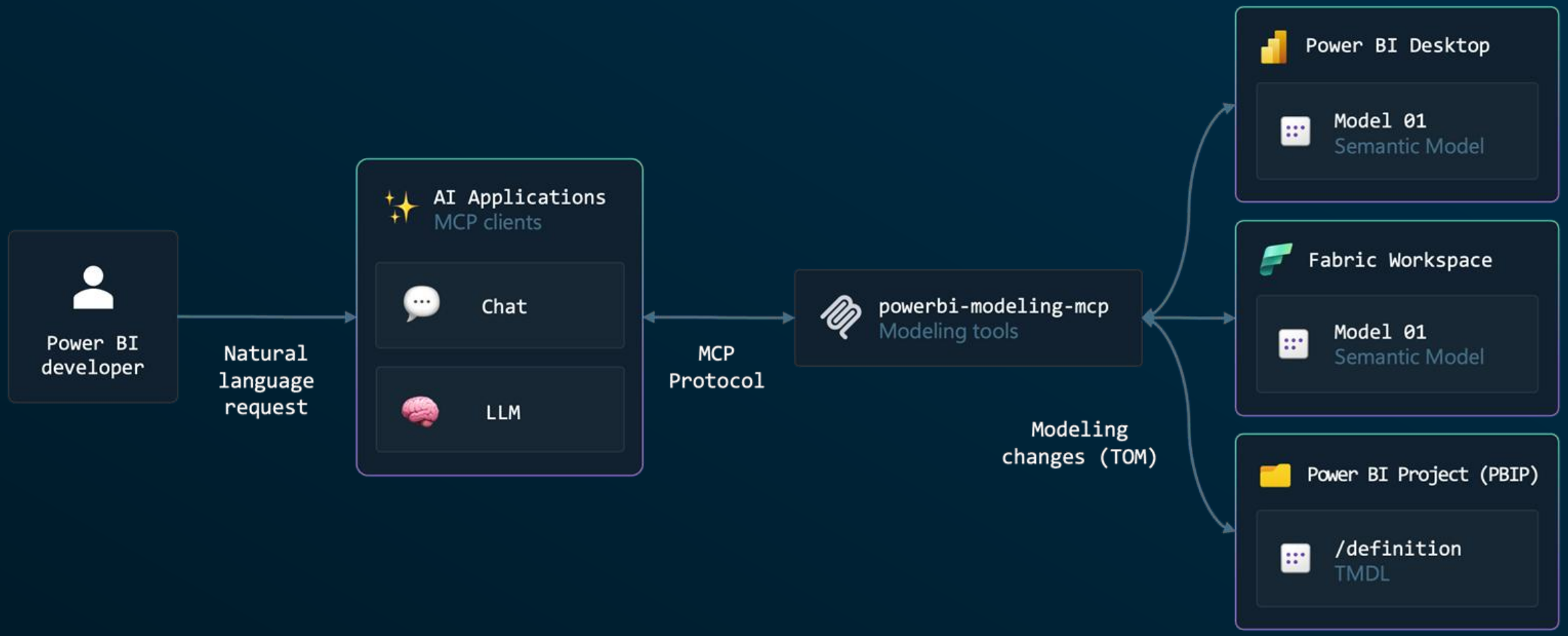
- ⚙️ Generate or scaffold Fabric resource definitions
- 📑 Retrieve official OpenAPI specs and JSON schema for validation and code generation
- 📄 Get example request/response payloads to accelerate integration
- ✅ Query curated best-practice guidance (pagination, LROs, authentication patterns)

DEMO

Microsoft Fabric MCP Server

Power BI Modeling MCP Server

Architecture



Power BI Modeling MCP Server

Use Cases

🔄 Build and modify semantic models with natural language

⚡ Bulk operations at scale

✅ Apply modeling best practices

🤖 Agentic development workflows

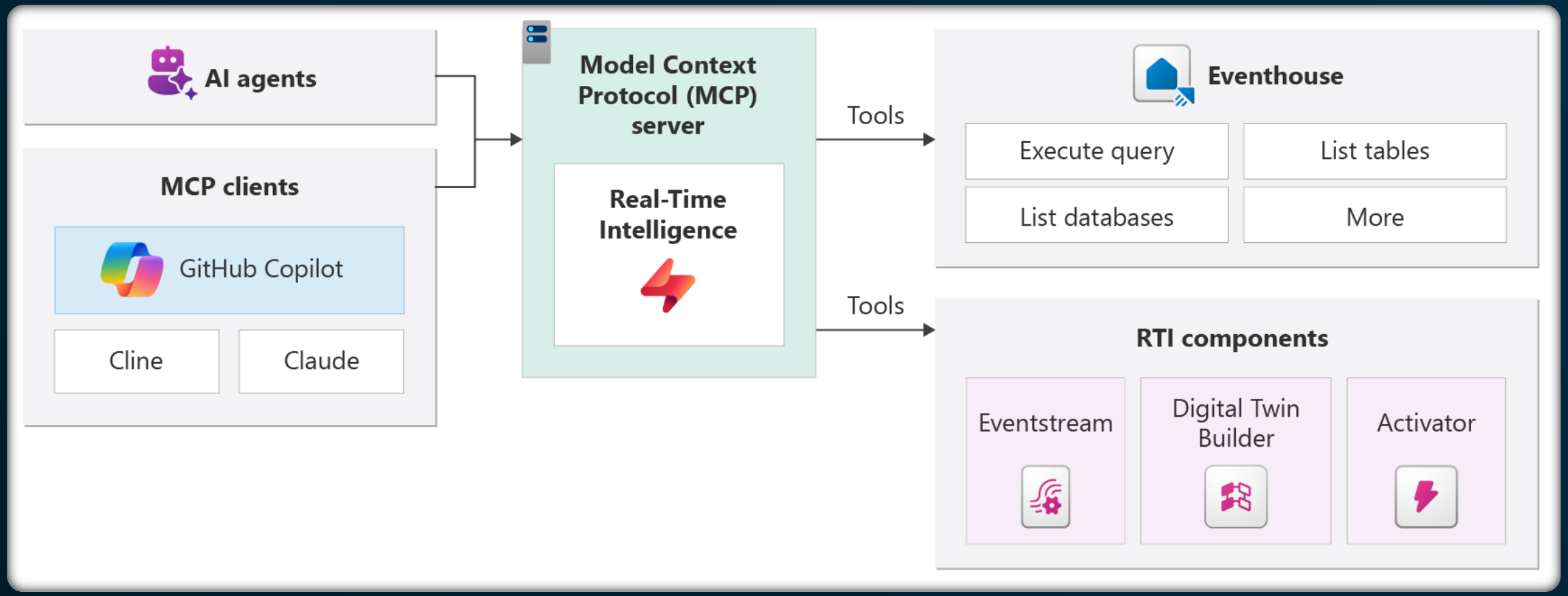
🔍 Query and validate DAX

DEMO

Power BI Modeling MCP Server

Fabric RTI MCP Server

Architecture



Fabric RTI MCP Server

Use Cases

Eventhouse: Execute KQL queries against Fabric Eventhouse and ADX

Eventstreams: Manage Eventstreams for real-time data processing

- List Eventstreams in workspaces
- Get Eventstream details and definitions
- Create new Eventstreams
- Update existing Eventstreams
- Delete Eventstreams

Activator: Create and manage Activator triggers for real-time alerting

- Create new triggers with KQL source monitoring
- Set up email and Teams notifications when a condition occurs
- List Activator artifacts in workspaces

DEMO

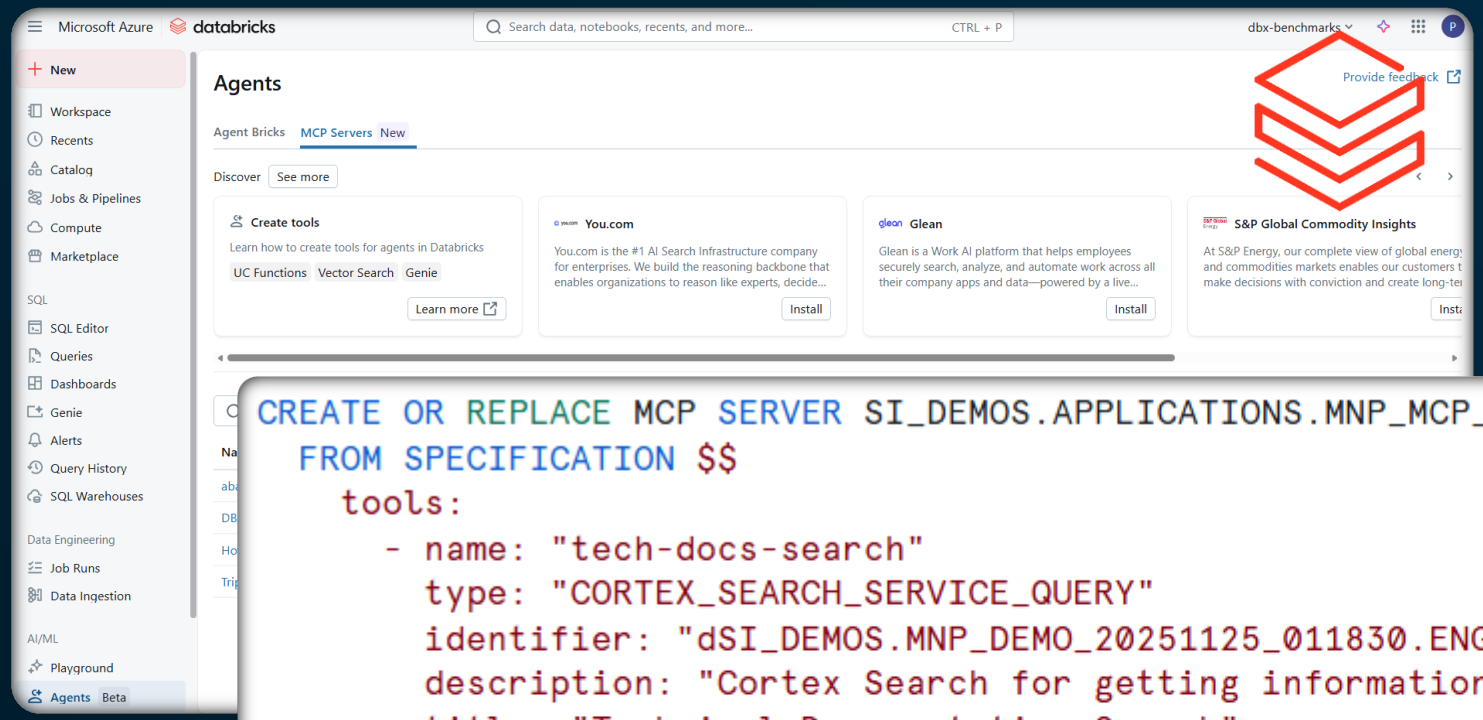
Fabric RTI MCP Server

Microsoft Data Factory MCP Server

Use Cases

- ⚙️ Manage Fabric gateways (including VNet gateways)
- 📄 List, retrieve and create connections in Fabric
- ✅ Manage pipelines, copy jobs and dataflows

How Fabric Compares in MCP vs Competitors?




```
CREATE OR REPLACE MCP SERVER SI_DEMOS.APPLICATIONS.MNP_MCP_SERVER
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      - name: "tech-docs-search"
        type: "CORTEX_SEARCH_SERVICE_QUERY"
        identifier: "dSI_DEMOS.MNP_DEMO_20251125_011830.ENGINE_TECHNICAL_DOCS_CHUNKS_SEARCH_SERVICE"
        description: "Cortex Search for getting information from technical documentation."
        title: "Technical Documentation Search"

      - name: "ticket-analytics-view"
        type: "CORTEX_ANALYST_MESSAGE"
        identifier: "SI_DEMOS.MNP_DEMO_20251125_011830.NEW_SEMANTIC_VIEW_SEMANTIC_MODEL"
        description: "Semantic view for JIRA ticket and SAP order data."
        title: "Semantic view for tickets and orders"
  $$
```

MCP Server Watchout List

⚠️ **Make sure you trust the LLM (MCP host)**
Are you OK with the LLM reading your data and/or metadata?

⚠️ **Don’t fully rely on AI**
It’s still LLM that does the job and it can hallucinate or misinterpret your prompts!

⚠️ **Understand available tools**
Your MCP server will be as good as the set of tools it provides.

⚠️ **Set up security beforehand**
Make sure you control data access and allowed commands before letting MCP tools in.

⚠️ **Never trust a 3rd party MCP**
MCP servers can contain harmful tools. Make sure you get them from trusted sources.

⚠️ **Watch out for costs and resource consumption**
Your interactions with MCP servers will consume tokens, Fabric CUs and other resources.

MCP and MCP for Fabric

Useful resources

- **MCP for beginners**
  https://github.com/microsoft/mcp-for-beginners
- **Microsoft MCP servers**
  https://github.com/microsoft/mcp
- **Microsoft Fabric MCP Server**
  https://github.com/microsoft/mcp/tree/main/servers/Fabric.Mcp.Server
- **MCP GraphQL Server**
  https://github.com/microsoft/fabric-samples/tree/main/docs-samples/data-engineering/GraphQL/MCP
- **Power BI Modeling MCP server**
  https://marketplace.visualstudio.com/items?itemName=analysis-services.powerbi-modeling-mcp
  https://github.com/microsoft/powerbi-modeling-mcp
- **Microsoft Data Factory MCP Server**
  https://github.com/microsoft/DataFactory.MCP
- **Fabric RTI MCP Server**
  https://github.com/microsoft/fabric-rti-mcp/
- **Semantic Model MCP Server (Fabric CAT)**
  https://github.com/microsoft/fabric-toolbox/tree/main/tools/SemanticModelMCPServer

MCP and MCP for Fabric

Useful resources from Fabric community

- **Kurt Buhler: Control Power BI and Fabric by using AI tools (with MCP servers)**
  https://www.youtube.com/watch?v=03WjmMKooFM
- **Kurt Buhler: Learn MCP Servers & AI for Data and Analytics (playlist)**
  https://www.youtube.com/playlist?list=PLXa38gbQxdE1WM3RuU02v_4timu7dnIXr
- **Guy in a Cube: Chat with your Eventhouse using Fabric's MCP Server**
  https://www.youtube.com/watch?v=vMyh4S8zrZY
- **Injae Park: Official Power BI MCP - Quick Start Guide [Nov 2025]**
  https://www.youtube.com/watch?v=0ZRDTZm1iaI
- **Kurt Buhler: Agentic orchestration of Fabric by using the Fabric CLI (not MCP but super useful)**
  https://tabulareditor.com/blog/agentic-orchestration-of-fabric-by-using-the-fabric-cli

⬇️ Download this presentation from **GitHub.com/pawelpo/presentations**